SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT- 3

1. Student Details

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Division	A
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Date of Submission	15-09-2026

2. Practical Details

Practical No.	3
Practical Title	Class, Constructor, Destructor and User-Defined Functions
Date Performed	15-09-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Writeaprogramon operatoroverloading using default and parameterized constructors.
The program should demonstrate the use of a default constructor, parameterized constructor, and operator overloading in C++. The program should create Book objects, initialize their data using constructors, overload the +operator to add the prices and combine the names of two books, and display the resulting book information.

4. Objectives

1.To understand the concept of operator overloading in C++.
2.To implement default and parameterized constructors.
3.To create and use objects of a class.
4.To overload the + operator for user-defined objects.
5.To understand how constructors initialize object data.

5. Expected Learning Outcomes

- Explain the concept of operator overloading in C++.
- Implement default and parameterized constructors in a C++ program.
- Define and use an overloaded operator for class objects.
- Understand how data members are initialized and accessed through objects.
- Display the result of an operation performed on user-defined objects.

6. Theory / Concept

Operator Overloading: It allows operators such as +, -, *, etc. to work with user-defined objects.
Default Constructor: A constructor without parameters. It initializes the object with default values.
Parameterized Constructor: A constructor that accepts parameters and initializes an object with given values.
In this program, the + operator is overloaded to add the price and names of two Book objects.

7. Algorithm / Pseudocode

START

Step 1: Create a Book class with price and name.
Step 2: Create a default constructor to initialize price and name to zero/empty values.
Step 3: Create a parameterized constructor to initialize book details.
Step 4: Overload the + operator to add two Book objects.
Step 5: Create objects, perform addition using b2 + b3, and display the result. STOP

08. Technology / Tools / Libraries Used

Technology / Tool C++	**Purpose**
IDE / Platform	Programming Language
Dev-C++ / Code::Blocks	IDE / Platform
iostream	Input and Output

09. Implementation / GitHub Repository

Particular	**Details**
GitHub Repository URL	
Repository Name	**Assigenment3**
Branch Used	**main**
Commit ID / Latest Commit	**Add file via upload**
Particular	**Details**
Date of Last Commit	**15-09-2026**

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
1	Book b2(500, "C++");	Book Name: C++ Total Price: 500	Book Name: C++ Total Price: 500	Pass
2	Book b3(300, "Python");	Book Name: Python Total Price: 300	Book Name: Python Total Price: 300	Pass
3	b1 = b2 + b3;	Book Name: C++ & Python Total Price: 800	Book Name: C++ & Python Total Price: 800	Pass

```
PS C:\Users\Homesh\Desktop\20240110075_Project files> cd "c:\Users\Homesh" ; if ($?) { g++ opps.cpp -o opps } ; if ($?) { .\opps
{ .\opps }
First Book:
Book Name: C++
Total Price: 500

Second Book:
Book Name: Python
Total Price: 300

After Operator Overloading:
Book Name: C++ & Python
Total Price: 800

PS C:\Users\Homesh\Desktop\20240110075_Project files>
```

11.Results / Observations

The C++ program was implemented successfully in Visual Studio Code. The Book class was created with price and name data members. The default constructor initialized an object with default values, while the parameterized constructor initialized the book objects with given values.

The + operator was successfully overloaded to add the prices and combine the names of two Book objects. The resulting object was displayed using the display() member function.

12. Conclusion

Thepractical successfully demonstrated operator overloading using default and parameterized constructors in C++. It helped in understanding how constructors initialize objects and how an overloaded operator can be used to perform operations on user-defined class objects.

13. Student Declaration

Ihereby declare that theabovepractical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Homesh Sawarkar	202501110069	Homesh.A.S